LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Noise suppression and its levels
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Noise suppression and its levels
5. Specific Objectives	13.1.4.1 Trainee will be able to describe various methods of noise suppression.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of noise suppression diagrams in IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary source of noise in a gas turbine engine?
(b) Define the term 'noise suppression' in the context of aviation.
(c) Why is noise control critical for aircrew and ground personnel?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to engine noise sources • Decibel scale and noise levels • Active noise control methods • Passive noise suppression techniques • Regulatory noise limits	L & Ds	PPT, IB	15 min
(b)	• Analysis of inlet and exhaust noise • Fan blade noise reduction strategies • Liner materials and acoustic properties • Chequerboard and honeycomb liners • Impact of noise on crew health	L & Ds	PPT, PDF	20 min
(c)	• Comparison of active vs passive systems • Future trends in noise suppression • Maintenance requirements for liners • Case study of modern engine noise • Summary of suppression levels	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between active and passive noise suppression methods.
(b) Describe how chequerboard liners contribute to noise reduction.
(c) Identify two factors that influence the effectiveness of noise suppression liners.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of noise suppression discussed in this lesson apply to the thrust reversal systems covered in LP 13.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various methods of thrust reversal and principles of operation.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min